

38.2 Quasi-likelihood estimating equations

Estimation here starts from Eq. 38.1.1 and not from a density. There are two useful ways to read it: as the derivative of a function that is not quite a log-likelihood but close enough to carry the usual arguments, and as the best member of a large family of estimating equations. This section develops both, and they meet in the middle.

38.2.1 The quasi-likelihood function

Definition 38.2.1 — Quasi-likelihood

Under Definition 38.1.1, the **quasi-likelihood** of a mean $\mu \in \mathcal{M}$ given a response y in the closure of $\mathcal{M}$, with weight w , is

$$Q(\mu; y) = \frac{w}{\phi} \int_y^\mu \frac{y-t}{V(t)} dt, \quad (38.2.1)$$

assumed finite, the integral being read as a limit at an endpoint of $\mathcal{M}$ where it converges — as it does at $y = 0$ for $V(t) = t$ and at $y \in \{0, 1\}$ for $V(t) = t(1-t)$. The quasi-likelihood of the model is $Q(\boldsymbol{\beta}) = \sum_{i=1}^n Q(\mu_i; y_i)$ with $\mu_i = h(\mathbf{x}_i^\top \boldsymbol{\beta})$ and weight w_i .

The definition is designed to make one identity true by construction, and everything else follows from it.

Proposition 38.2.1 — What the quasi-likelihood does and does not share with a log-likelihood

Write $Q' = \partial Q / \partial \mu$. Then, for every variance specification:

- (a) $Q'(\mu; y) = w(y - \mu) / \{\phi V(\mu)\}$, and $Q(\mu; y) \leq 0$ with equality only at $\mu = y$: the quasi-likelihood is maximized at the observation, as a saturated fit would be;
- (b) $E\{Q'(\mu; Y)\} = 0$ and $\text{Var}\{Q'(\mu; Y)\} = -E\{\partial^2 Q / \partial \mu^2\} = w / \{\phi V(\mu)\}$, the two Bartlett identities;
- (c) if (V, ϕ, w) are those of an exponential dispersion family (Definition 34.1.1), then $Q(\mu; y) = \ell(\mu; y) - \ell(y; y)$, so Q is the log-likelihood measured from its saturated value, and $-2\phi Q(\boldsymbol{\beta}) = D(\mathbf{y}, \boldsymbol{\mu})$ is the deviance of Eq. 34.5.4.

Proof.

- (a) Differentiating Eq. 38.2.1 in its upper limit gives the stated Q' . Because $V > 0$, the integrand $(y-t)/V(t)$ is positive for $t < y$ and negative for $t > y$, so for $\mu > y$ the integral is of a negative function and for $\mu < y$ it is minus the integral of a positive one: both give $Q \leq 0$, with equality only for an empty interval.
- (b) $E(Y - \mu) = 0$ gives the first identity, and $\text{Var}(Y) = \phi V(\mu)/w$ gives $\text{Var}(Q') =$

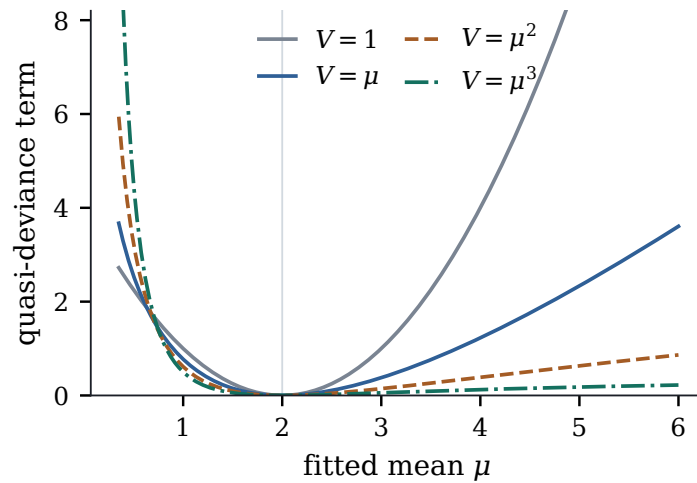


Figure 38.2.1: One term of the quasi-deviance Eq. 38.2.2 at $y = 2$, $w = 1$, against the fitted mean, for four power variance functions. Only the constant one penalizes symmetrically.

$w/\{\phi V(\mu)\}$. Differentiating Q' again,

$$\frac{\partial^2 Q}{\partial \mu^2} = -\frac{w}{\phi} \cdot \frac{V(\mu) + (y - \mu)V'(\mu)}{V(\mu)^2},$$

whose expectation is $-w/\{\phi V(\mu)\}$, since the term carrying $y - \mu$ has mean zero.

- (c) For an exponential dispersion family the chain rule used in the proof of Theorem 34.3.1 gives $\partial \ell / \partial \mu = w(y - \mu) / \{\phi V(\mu)\}$, which is Q' ; two functions of μ with the same derivative differ by a constant, and both sides vanish at $\mu = y$. Comparing with Eq. 34.5.4 gives $D = -2\phi Q$.

■

The converse of (c) fails: $\exp Q$ is in general not a density in y (Exercise (C1)). Part (c) is nevertheless why the objects here look familiar — everything the exponential family contributed to Definition 34.5.1 is already in the integral Eq. 38.2.1 — and for a variance function from no family Eq. 38.2.1 still measures the discrepancy between a fit and an observation, the **quasi-deviance**

$$D(\mathbf{y}, \boldsymbol{\mu}) = -2\phi Q(\boldsymbol{\beta}) = 2 \sum_{i=1}^n w_i \int_{\mu_i}^{y_i} \frac{y_i - t}{V(t)} dt \geq 0, \quad (38.2.2)$$

which does not involve ϕ and vanishes only at $\boldsymbol{\mu} = \mathbf{y}$. Figure 38.2.1 draws one term for four power variance functions: only $V \equiv 1$ penalizes symmetrically, and under $V(\mu) = \mu$ a fitted mean one unit too small costs 2.043 times as much as one unit too large, against 4.254 for $V(\mu) = \mu^2$ and 9.000 for $V(\mu) = \mu^3$. A variance function that grows with the mean is a statement about which errors matter.

```
import numpy as np
from scipy import integrate

def quasi_deviance_term(y, mu, V):
    """2 * integral from mu to y of (y - t)/V(t) dt: one observation's quasi-
    deviance."""
```

```

value, _ = integrate.quad(lambda t: (y - t) / V(t), mu, y)
return 2 * value

y0 = 2.0
closed_form = {
    "V = 1": (lambda m: (y0 - m) ** 2, lambda t: 1.0),
    "V = mu": (lambda m: 2 * (y0 * np.log(y0 / m) - (y0 - m)), lambda t: t),
    "V = mu^2": (lambda m: 2 * (-np.log(y0 / m) + (y0 - m) / m), lambda t: t ** 2),
    "V = mu^3": (lambda m: (y0 - m) ** 2 / (y0 * m ** 2), lambda t: t ** 3),
}
for name, (formula, V) in closed_form.items():
    print(f"{name:9s} at mu = 0.8: {quasi_deviance_term(y0, 0.8, V):.6f}"
          f" closed form {formula(0.8):.6f}")

```

[Run this code online](#)

38.2.2 The quasi-score

Theorem 38.2.2 — The quasi-score

Let $\mathbf{U}(\boldsymbol{\beta}) = \partial Q / \partial \boldsymbol{\beta}$ be the **quasi-score**, and keep the notation $\mathbf{W} = \text{diag}\{w_i h'(\eta_i)^2 / V(\mu_i)\}$ and $\mathbf{D} = \text{diag}\{h'(\eta_i)\}$ of Theorem 34.3.1. Write $\mathbf{A}_n = \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$. Then:

- (a) $\mathbf{U}(\boldsymbol{\beta}) = \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu})$, which is Eq. 34.3.1 exactly. The estimating equations $\mathbf{U}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$ are Eq. 38.1.1, they do not involve ϕ , and they are solved by the iteratively reweighted least squares of Theorem 34.4.1;
- (b) if the mean model is correct then $E \mathbf{U}(\boldsymbol{\beta}^0) = \mathbf{0}$, *whatever the true variances are*: the quasi-score is an unbiased estimating function;
- (c) $-E\{\partial \mathbf{U} / \partial \boldsymbol{\beta}^\top\} = \mathbf{A}_n$, again whatever the true variances are;
- (d) if in addition the variance specification Eq. 38.1.2 is correct, then $\text{Cov}\{\mathbf{U}(\boldsymbol{\beta}^0)\} = \mathbf{A}_n$ as well: sensitivity and variability agree, which is the second Bartlett identity for Q ;
- (e) if instead $\text{Var}(Y_i) = \sigma_i^2$ for arbitrary σ_i^2 , then

$$\text{Cov}\{\mathbf{U}(\boldsymbol{\beta}^0)\} = \mathbf{B}_n = \frac{1}{\phi^2} \sum_{i=1}^n \left\{ \frac{w_i h'(\eta_i)}{V(\mu_i)} \right\}^2 \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top, \quad (38.2.3)$$

which equals $\mathbf{A}_n$ if and only if $\sum_i \{w_i h'_i / V_i\}^2 \{\sigma_i^2 - \phi V_i / w_i\} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top = \mathbf{0}$.

Proof.

- (a) By Proposition 38.2.1(a) and the chain rule,

$$\frac{\partial Q}{\partial \beta_j} = \sum_{i=1}^n Q'(\mu_i; y_i) \frac{d\mu_i}{d\eta_i} x_{ij} = \frac{1}{\phi} \sum_{i=1}^n x_{ij} \frac{w_i h'(\eta_i)}{V(\mu_i)} (y_i - \mu_i),$$

which is the coordinate form of Eq. 34.3.1. The map from $(V, h, \mathbf{w}, \phi)$ to it is the one computed in Theorem 34.3.1(a), so the matrix form, the absence of ϕ and the iteration of Theorem 34.4.1 transfer unchanged; none of the three used the density.

- (b) Every term carries the factor $y_i - \mu_i$, which has mean zero when $\mu_i = h(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^0)$. No assumption about $\text{Var}(Y_i)$ is used.
- (c) Write $a(\eta) = h'(\eta)/V\{h(\eta)\}$, so that $\mathbf{U} = \phi^{-1} \sum_i w_i a(\eta_i)(y_i - \mu_i) \mathbf{x}_{(i)}$. Differentiating,

$$\frac{\partial \mathbf{U}}{\partial \boldsymbol{\beta}^\top} = \frac{1}{\phi} \sum_{i=1}^n w_i \{a'(\eta_i)(y_i - \mu_i) - a(\eta_i)h'(\eta_i)\} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top,$$

the computation of Theorem 34.3.1(c). Expectation annihilates the first term whatever the variances, and $w_i a(\eta_i) h'(\eta_i) = \mathbf{W}_{ii}$ leaves $-\mathbf{A}_n$.

- (d) and (e). The summands of $\mathbf{U}$ are independent with mean zero, so $\text{Cov}(\mathbf{U}) = \phi^{-2} \sum_i w_i^2 a(\eta_i)^2 \text{Var}(Y_i) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$, which is Eq. 38.2.3. Substituting $\sigma_i^2 = \phi V(\mu_i)/w_i$ turns the i th coefficient into $\phi^{-2} w_i^2 \{h'_i/V_i\}^2 \phi V_i/w_i = \phi^{-1} w_i h_i'^2/V_i$, giving $\mathbf{A}_n$. The stated condition is the difference of the two sums.

■

Remark (Existence and uniqueness). Nothing above guarantees that Eq. 38.1.1 has a solution, or only one. Under the canonical link *of the variance function*, $g'(\mu) = 1/V(\mu)$, the proof of (c) has $a \equiv 1$ and $a' \equiv 0$, so the Jacobian is $-\mathbf{A}_n$, negative definite whenever $\text{rank}(\mathbf{X}) = p$; Q is then strictly concave and a root, if one exists, is the unique maximizer, by Theorem 34.3.3 with b'' replaced by V . For other links Q can have several stationary points, as the warning after Theorem 34.3.3 showed.

Example. (Strike durations)

Kennan's data record the length in days of 62 strikes in US manufacturing, with a measure of unanticipated industrial production at the time each strike began. Take a log link and $V(\mu) = \mu$. The production coefficient is -7.680 ; replacing $V(\mu) = \mu$ by $V(\mu) = 7.5\mu$ returns the same number to ten decimal places, since ϕ cancels from Eq. 38.1.1, while $V(\mu) = \mu^2$ moves it to -9.353 , a quadratic variance function downweighting the long strikes.

The mean duration is 42.66 days and the sample variance 2102.7, nearly fifty times as large: a Poisson likelihood is no serious description of these data, and the question is how much of the Poisson machinery survives anyway.

```
import numpy as np
import statsmodels.api as sm

data = sm.datasets.strikes.load_pandas().data
y = data["duration"].to_numpy(float)          # length of the strike, in days
x = data["iprod"].to_numpy(float)             # unanticipated industrial production
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape

def quasi_irls(X, y, V, steps=60):
    """Solve the quasi-score equations for a log link and variance function V."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    for _ in range(steps):
        eta = X @ beta
        mu = np.exp(eta)
```

```

W = mu ** 2 / V(mu)                # working weight (dmu/deta)^2 / V(mu)
z = eta + (y - mu) / mu            # working response
beta = np.linalg.solve((X.T * W) @ X, (X.T * W) @ z)
return beta

beta_lin = quasi_irls(X, y, lambda m: m)          # V(mu) = mu
beta_scaled = quasi_irls(X, y, lambda m: 7.5 * m) # V(mu) = phi mu: same
equations
beta_sq = quasi_irls(X, y, lambda m: m ** 2)      # V(mu) = mu^2: different
equations
print(f"V = mu      {beta_lin}")
print(f"V = 7.5 mu  {beta_scaled}")
print(f"V = mu^2    {beta_sq}")

```

[Run this code online](#)

38.2.3 Optimality among linear estimating functions

The second view asks what else might have been used. Any function of the data and the parameter with mean zero at β^0 is an **unbiased estimating function**, and setting it to zero defines an estimator; the quasi-score is the best of a natural class.

Proposition 38.2.3 — Godambe optimality

Assume Eq. 38.1.2, and consider **linear estimating functions**

$$\mathbf{g}(\beta) = \sum_{i=1}^n \mathbf{a}_i(\beta) \{y_i - \mu_i(\beta)\} = \mathbf{C}(\beta) (\mathbf{y} - \boldsymbol{\mu}),$$

where $\mathbf{C} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ is $p \times n$, differentiable in β , and does not depend on $\mathbf{y}$. Let $\mathbf{S} = -E\{\partial \mathbf{g} / \partial \beta^\top\}$ and suppose $\mathbf{S}$ is nonsingular. Define the **Godambe information**

$$\mathcal{J}(\mathbf{g}) = \mathbf{S}^\top \text{Cov}\{\mathbf{g}(\beta^0)\}^{-1} \mathbf{S}.$$

- (a) Every such $\mathbf{g}$ is unbiased, and $\mathbf{S} = \mathbf{C} \mathbf{D} \mathbf{X}$.
- (b) $\mathcal{J}(\mathbf{g}) \preceq \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X} = \mathcal{J}(\mathbf{U})$ in the order of nonnegative definiteness, for every linear estimating function $\mathbf{g}$.
- (c) Equality holds if and only if $\mathbf{C}^\top = \boldsymbol{\Sigma}^{-1} \mathbf{D} \mathbf{X} \mathbf{N}$ for a nonsingular $p \times p$ matrix $\mathbf{N}$, with $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$: the coefficients are those of the quasi-score up to one common nonsingular transformation, and $\mathbf{g}$ then has the same roots as $\mathbf{U}$.

Proof.

- (a) Unbiasedness follows from $E(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$. Differentiating,

$$\frac{\partial \mathbf{g}}{\partial \beta^\top} = \sum_{i=1}^n \frac{\partial \mathbf{a}_i}{\partial \beta^\top} (y_i - \mu_i) - \mathbf{C} \frac{\partial \boldsymbol{\mu}}{\partial \beta^\top},$$

read entry by entry. Its expectation vanishes because $\mathbf{a}_i$ does not depend on $\mathbf{y}$,

and $\partial\boldsymbol{\mu}/\partial\boldsymbol{\beta}^\top = \mathbf{D}\mathbf{X}$, so $\mathbf{S} = \mathbf{C}\mathbf{D}\mathbf{X}$.

(b) and (c). Under Eq. 38.1.2, $\boldsymbol{\Sigma} = \text{diag}\{\phi V(\mu_i)/w_i\}$ is positive definite. Set

$$\mathbf{F} = \boldsymbol{\Sigma}^{1/2}\mathbf{C}^\top, \quad \mathbf{K} = \boldsymbol{\Sigma}^{-1/2}\mathbf{D}\mathbf{X},$$

both $n \times p$. Then $\text{Cov}(\mathbf{g}) = \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top = \mathbf{F}^\top\mathbf{F}$ and $\mathbf{S} = \mathbf{C}\mathbf{D}\mathbf{X} = \mathbf{F}^\top\mathbf{K}$, so

$$\mathcal{J}(\mathbf{g}) = \mathbf{K}^\top\mathbf{F}(\mathbf{F}^\top\mathbf{F})^{-1}\mathbf{F}^\top\mathbf{K} = \mathbf{K}^\top\mathbf{P}\mathbf{K},$$

where $\mathbf{P} = \mathbf{F}(\mathbf{F}^\top\mathbf{F})^{-1}\mathbf{F}^\top$ is the orthogonal projection onto $\mathcal{C}(\mathbf{F})$ (Theorem 6.3.7), and $\mathbf{F}^\top\mathbf{F}$ is nonsingular because $\mathbf{S}$ is. As $\mathbf{I} - \mathbf{P}$ is symmetric idempotent, $\mathbf{v}^\top(\mathbf{I} - \mathbf{P})\mathbf{v} = \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2 \geq 0$, so $\mathbf{P} \preceq \mathbf{I}$ and $\mathcal{J}(\mathbf{g}) \preceq \mathbf{K}^\top\mathbf{K}$. Finally

$$\mathbf{K}^\top\mathbf{K} = \mathbf{X}^\top\mathbf{D}\boldsymbol{\Sigma}^{-1}\mathbf{D}\mathbf{X} = \sum_{i=1}^n \frac{w_i h'(\eta_i)^2}{\phi V(\mu_i)} \mathbf{x}_{(i)}\mathbf{x}_{(i)}^\top = \frac{1}{\phi} \mathbf{X}^\top\mathbf{W}\mathbf{X},$$

and Theorem 38.2.2(a),(d) identify this as $\mathcal{J}(\mathbf{U})$, since for the quasi-score sensitivity and variability are both $\mathbf{A}_n$.

Equality in (b) means $\mathbf{K}^\top(\mathbf{I} - \mathbf{P})\mathbf{K} = \mathbf{0}$, that is $\mathcal{C}(\mathbf{K}) \subseteq \mathcal{C}(\mathbf{F})$. Both matrices have p columns and, because $\mathbf{F}^\top\mathbf{K}$ is nonsingular, rank p ; so their column spaces coincide and $\mathbf{K} = \mathbf{F}\mathbf{N}$ for a nonsingular $\mathbf{N}$. Written out this is $\mathbf{C}^\top = \boldsymbol{\Sigma}^{-1}\mathbf{D}\mathbf{X}\mathbf{N}^{-1}$, whose i th row is a fixed multiple of $w_i h'(\eta_i)\mathbf{x}_{(i)}^\top/V(\mu_i)$. Then $\mathbf{g} = \mathbf{N}^{-\top}\mathbf{U}$, because $\mathbf{X}^\top\mathbf{D}\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$ already carries the factor ϕ^{-1} that $\mathbf{U}$ has; so $\mathbf{g}$ vanishes exactly where $\mathbf{U}$ does. ■

Warning. The optimality is *within the class of linear estimating functions*. A known true density would have a score nonlinear in $\mathbf{y}$ with a larger Godambe information — the NB1 score Eq. 37.4.6 is an instance — so “best linear” is the honest reading, as in Theorem 7.2.1. Optimality also assumes Eq. 38.1.2; under a wrong variance function the quasi-score stays unbiased but is no longer optimal, and Section 38.3 prices the loss.

38.2.4 Exercises

38.2.4.1 A. Check your understanding

Exercise A1.

Compute $\phi w^{-1}Q(\mu; y)$ from Eq. 38.2.1 for $V(\mu) = 1$, $V(\mu) = \mu$, $V(\mu) = \mu(1 - \mu)$ and $V(\mu) = \mu^2$, and check that in each case it is the log-likelihood kernel of the corresponding family, up to a function of y alone.

Solution. For $V = 1$, $\int_y^\mu (y - t) dt = -(y - \mu)^2/2$. For $V(t) = t$ the integrand is $y/t - 1$, giving the Poisson kernel $y \log \mu - \mu$. For $V(t) = t(1 - t)$, partial fractions give $y/t - (1 - y)/(1 - t)$, whose integral is the binomial kernel $y \log \mu + (1 - y) \log(1 - \mu)$. For $V(t) = t^2$ the integrand is $y/t^2 - 1/t$, giving the gamma kernel $-y/\mu - \log \mu$.

Exercise A2.

Show that the NB2 variance function $V(\mu) = \mu + \mu^2/\kappa$ with κ known gives $\phi w^{-1}Q(\mu; y) = y \log\{\mu/(\mu + \kappa)\} + \kappa \log\{\kappa/(\mu + \kappa)\}$ up to a function of y , which is the negative binomial log-likelihood kernel of Eq. 37.4.1. NB2 with κ known is an exponential dispersion family, so Proposition 38.2.1(c) applies; explain why the *quasi-Poisson* specification $V(\mu) = \mu$ with ϕ free does not enjoy the same conclusion, and relate this to the remark of Section 38.1.

38.2.4.2 B. Practice**Exercise B1.**

Take n independent responses with $EY_i = \mu$ (one unknown parameter, no regressors) and $\text{Var } Y_i = \phi V(\mu)/w_i$. Show that the quasi-score estimate of Theorem 38.2.2 is the weighted mean $\sum_i w_i y_i / \sum_i w_i$, and verify Proposition 38.2.3 directly: among estimating functions $\sum_i a_i(y_i - \mu)$ with constants a_i , the Godambe information is maximized by $a_i \propto w_i$.

Solution. Parameterizing by μ makes $h' \equiv 1$, so Eq. 38.1.1 reads $\sum_i w_i(y_i - \mu)/V(\mu) = 0$ and $\hat{\mu} = \sum_i w_i y_i / \sum_i w_i$. For a general $\mathbf{a}$ the sensitivity is $S = \sum_i a_i$ and the variability is $\phi V(\mu) \sum_i a_i^2/w_i$, so

$$J = \frac{(\sum_i a_i)^2}{\phi V(\mu) \sum_i a_i^2/w_i} \leq \frac{\sum_i w_i}{\phi V(\mu)}$$

by the Cauchy–Schwarz inequality applied to $\sum_i a_i = \sum_i (a_i/\sqrt{w_i})\sqrt{w_i}$, with equality if and only if $a_i \propto w_i$. The bound is $\phi^{-1}\mathbf{X}^\top \mathbf{W} \mathbf{X}$ with $\mathbf{X} = \mathbf{1}$.

Exercise B2.

Suppose the link satisfies $g'(\mu) = 1/V(\mu)$, the canonical link of the variance function. Show that Eq. 38.1.1 becomes $\sum_i w_i x_{ij}(y_i - \hat{\mu}_i) = 0$ for every j , so that the fit reproduces the weighted column totals of $\mathbf{X}$ exactly, as in Corollary 34.3.2. Check the condition for $V(\mu) = \mu$ with a log link and for $V(\mu) = \mu(1 - \mu)$ with a logit link, and find the canonical link of $V(\mu) = \mu^2$.

Solution. Since h is the inverse of g , $h'(\eta) = 1/g'(\mu) = V(\mu)$, so the coefficient $w_i h'(\eta_i)/V(\mu_i)$ in Eq. 38.1.1 is w_i . For $V(\mu) = \mu$ the log link has $g'(\mu) = 1/\mu$; for $V(\mu) = \mu(1 - \mu)$ the logit link has $g'(\mu) = 1/\{\mu(1 - \mu)\}$; for $V(\mu) = \mu^2$ the condition $g'(\mu) = \mu^{-2}$ gives $g(\mu) = -1/\mu$, the canonical link of the gamma family. The conclusion is the quasi-likelihood form of Corollary 34.3.2, proved there for exponential dispersion families but needing only Eq. 38.1.1.

38.2.4.3 C. Going deeper**Exercise C1.**

Take the quasi-binomial specification $V(t) = t(1 - t)$ on $(0, 1)$ with $w = 1$ and any $\phi > 0$. (a) Show that $y \mapsto \exp\{Q(\mu; y)\}$ is integrable over $(0, 1)$, so normalizes to a density f .

(b) Show that a distribution on $[0, 1]$ with mean m has variance at most $m(1 - m)$, with equality only on $\{0, 1\}$; deduce that f has variance strictly below $m(1 - m)$ at its own mean m , never $\phi m(1 - m)$ with $\phi \geq 1$. (c) Show that f is symmetric about $1/2$ when $\mu = 1/2$, and check numerically that at $\mu = 0.3$, $\phi = 1$ it has mean 0.4408 and variance 0.0687, against $\mu = 0.3$ and $\phi\mu(1 - \mu) = 0.21$. Conclude that integrability of $\exp Q$ does not make a variance specification a family.

Solution.

- (a) $Q \leq 0$ by [Proposition 38.2.1\(a\)](#), so $\exp Q \leq 1$ on a bounded interval.
- (b) For $Y \in [0, 1]$, $E\{Y(1 - Y)\} \geq 0$ gives $EY^2 \leq m$ and hence $\text{Var } Y \leq m - m^2$; equality forces $Y(1 - Y) = 0$ almost surely, which a density on the open interval excludes.
- (c) By [Exercise \(A1\)](#) the exponent is $\phi^{-1}\{y \log \mu + (1 - y) \log(1 - \mu) - y \log y - (1 - y) \log(1 - y)\}$, unchanged by $y \mapsto 1 - y$ when $\mu = 1/2$. Elsewhere neither moment is reproduced: $\exp Q$ normalizes without the result being a distribution with the moments of Eq. 38.1.2.